

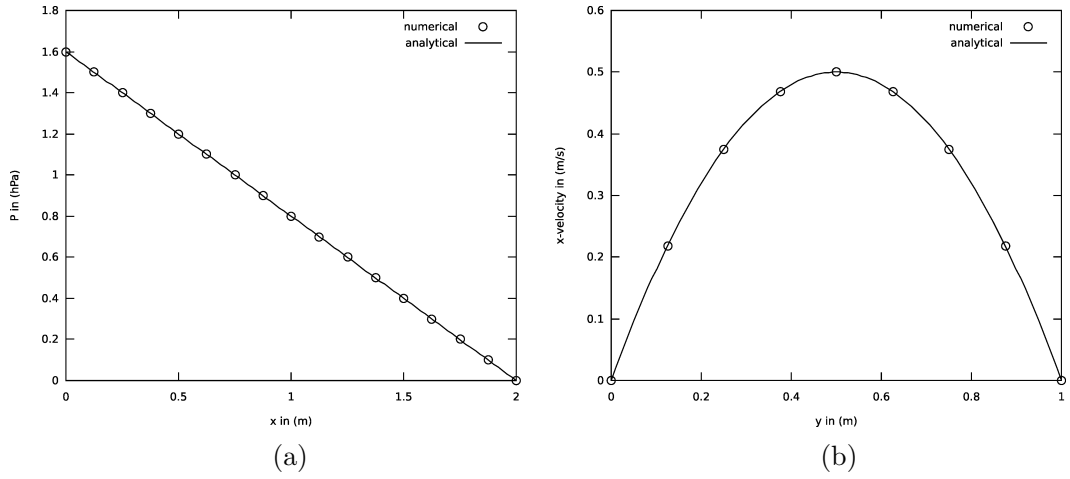


Figure 3.6: Numerical and analytic results of the channel flow (a) pressure in x -direction along $y = 0.5$ m (b) velocity v_x in y -direction along $x = 1$ m.

same as in the velocity driven case. At the in- and outlet the following pressure conditions are applied

$$\vec{h} = P\mathbf{1} \cdot \vec{n} \quad \text{with } P = 0.016 \text{ Pa} \quad \text{on } \Gamma_{\text{in}} \quad \text{and} \quad \vec{h} = 0 \quad \text{on } \Gamma_{\text{out}}, \quad (3.102)$$

while the x -velocity is free there. The comparison of the numerically computed x -velocity along the vertical line at $x = 1$ with the analytic solution is shown in Fig 3.6b. The spatial distribution of pressure and velocity is similar to the results depicted in Fig. 3.5. The pressure drop condition is of particular importance for phonation simulations as muscle contraction imposes a pressure drop along the glottis.

A good compliance with the analytical solution is confirmed with both types of boundary conditions.

Flow around a circular obstacle

A more complex and maybe the most famous benchmark in fluid mechanics is given by the flow around a cylinder. The list of literature about measurements and simulations of this setup is quite long; an overview can be found e.g. in [128]. The change from the steady to an unsteady flow around $Re = 50$ is noteworthy. The Kármán vortex street develops in the wake of the cylinder for Reynolds numbers above $Re = 50$. The air flow around an island may have similar flow structures, recognizable as soon as clouds exist, see Fig. 3.7. Two situations are discussed in detail: a steady ($Re = 40$) and an unsteady ($Re = 100$) flow. The geometry of the test case is chosen in accordance with [146] and is shown in Fig. 3.8 together with the finite element mesh. The mesh consist of 7876 Q2 elements.

Steady ($Re = 40$): The steady flow at $Re = 40$ around a cylinder is characterized by two symmetric recirculation eddies in the wake of the cylinder (see Fig. 3.9). The length of these eddies and the location of the flow separation point are famous benchmark results. The density and the dynamic viscosity of the fluid are set to

$$\rho = 1000 \text{ kg/m}^3 \quad \text{and} \quad \mu = 4 \times 10^{-4} \text{ Pa s}. \quad (3.103)$$

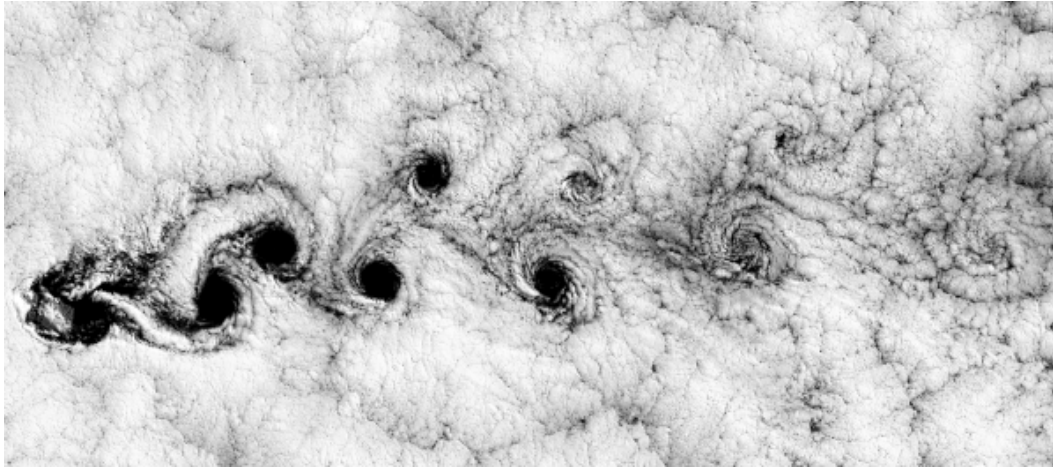


Figure 3.7: A Kármán vortex street around the island Guadalupe (picture from NASA www.nasa.com).

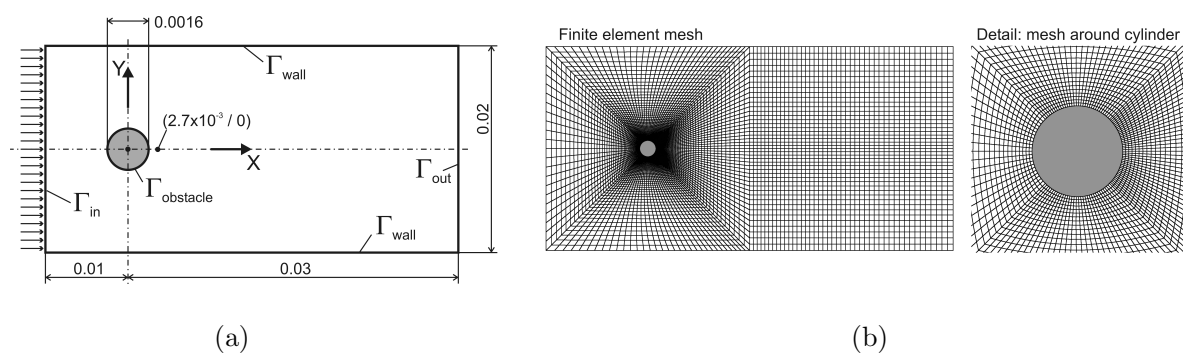


Figure 3.8: Flow around a cylinder. (a) Geometry of the fluid domain (units in m) (b) FEM mesh.

With a constant inflow velocity

$$v_x = 0.01 \text{ m/s} \quad \text{on } \Gamma_{\text{in}} \quad (3.104)$$

a Reynolds number of $Re = 40$ is obtained. A steady flow field develops [128] with the already mentioned two recirculation eddies (see Fig. 3.9) for $Re = 40$. Despite the steady behavior the numerical computation is performed with the unsteady algorithm, with a time step size of $\Delta t = 0.064 \text{ s}$, in order to verify that steady setups are resolved therewith. To compare the location of the flow separation point with published results, the angle α_r is introduced as well as the length of the recirculation zone L_r (see Fig. 3.9). The obtained

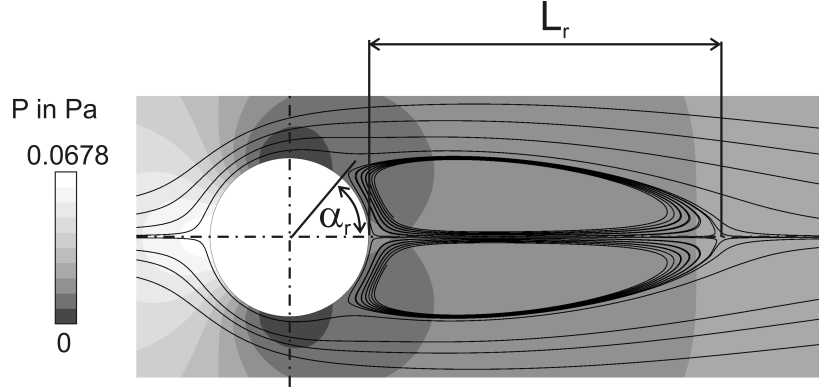


Figure 3.9: Streamlines on pressure for $Re = 40$ and the specification of the detaching angle α_r and the recirculation length L_r .

detaching angle $\alpha_r = 53^\circ$ and re-attaching length ratio $L_r/R = 4.29$ are both within the published range, see e.g. [146].

Further on, the pressure along the cylinder ($0^\circ \leq \alpha_r \leq 180^\circ$), see Fig. 3.10, is in compliance with the results of Choi [28]. To validate the pressure distribution, the pressure coefficient C_p is introduced

$$C_p = \frac{P - P_0 + \frac{1}{2}\rho||\vec{v}||^2}{\frac{1}{2}\rho||\vec{v}||^2}. \quad (3.105)$$

Unsteady ($Re = 100$): With a fluid density and dynamic viscosity of

$$\rho = 1000 \text{ kg/m}^3, \quad \mu = 1.6 \times 10^{-4} \text{ Pa s} \quad (3.106)$$

and a constant inflow velocity of

$$v_x = 0.01 \text{ m/s} \quad \text{on } \Gamma_{\text{in}} \quad (3.107)$$

a Reynolds number of $Re = 100$ is obtained and the Kármán vortex street develops in the wake of the cylinder. Now an unsteady flow field has to be captured for which the Strouhal number is an important measure. The time step size is thereby set to $\Delta t = 0.032 \text{ s}$. Restarting from a steady result, the Kármán vortex street develops and after 64 s a constant recurring eddy detaching exists. The unsteadiness of the flow is shown for ($64 \text{ s} \leq t \leq 67 \text{ s}$)

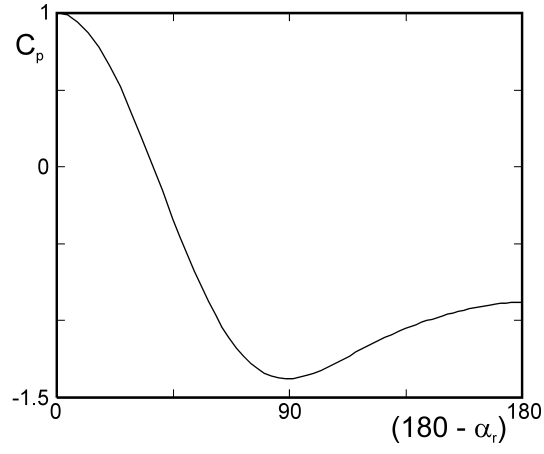


Figure 3.10: Pressure along the cylinder wall for $Re = 40$.

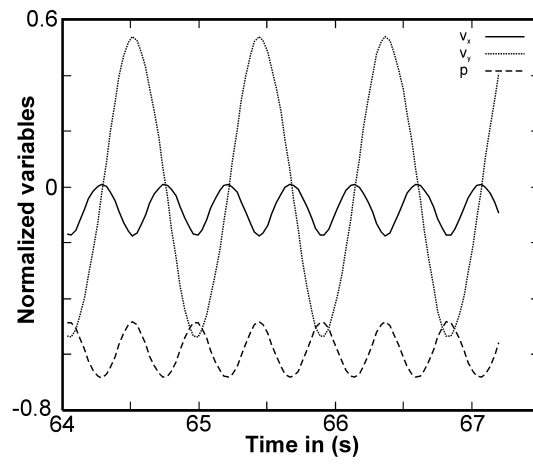
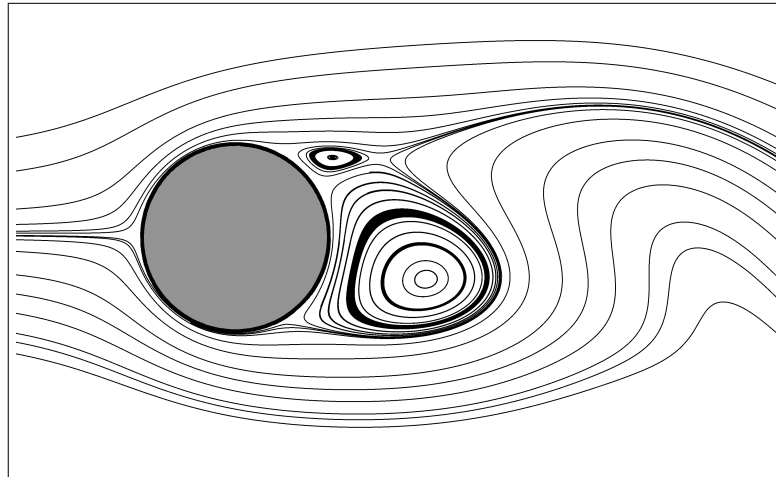
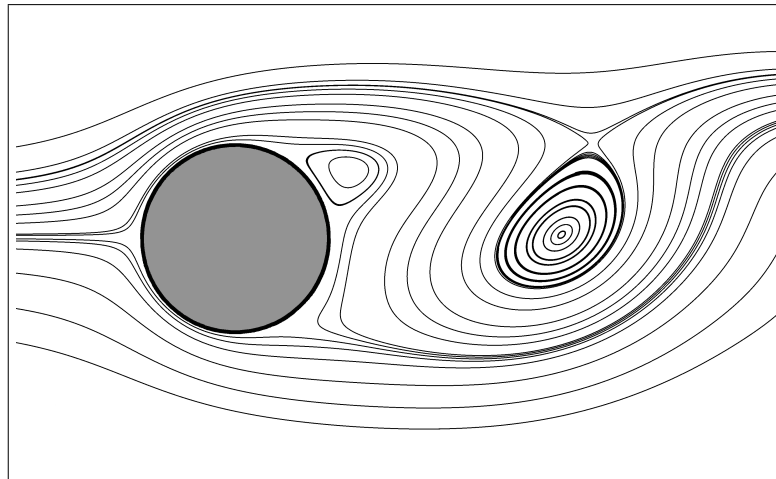


Figure 3.11: Normalized velocity and pressure at the point $(2.7 \times 10^{-3} \text{ m}/0 \text{ m})$ for $Re = 100$.

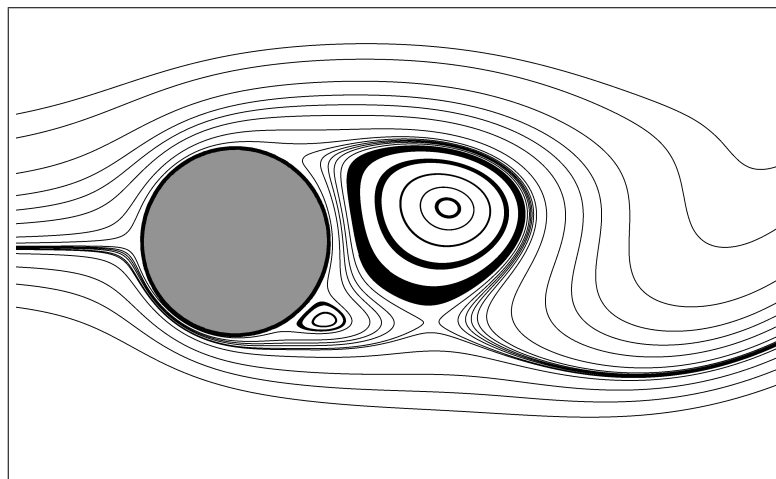
in Fig. 3.11 by the velocities and pressure values at the point $(2.7 \times 10^{-3} \text{ m}/0 \text{ m})$ in the wake of the cylinder. The obtained Strouhal number $St = 0.172$ is within the range of published results [146]. The periodic eddy detaching can be studied with the streamlines of three consecutive time steps capturing a half cycle, see Fig. 3.12. At t^1 the upper eddy has just appeared and the lower eddy has just detached. From t^1 to t^2 the lower eddy gets transported downstream and the upper eddy grows. In the time span from t^2 to t^3 the lower eddy moves further downstream and disappears. The upper eddy grows further. At the time step t^3 the upper eddy detaches and a new lower eddy appears. This cycle is repeated. The spatial distribution of velocity and pressure is displayed in Fig. 3.13 for the time step t^1 . Fig. 3.14 shows the velocity and pressure distribution along the horizontal centerline ($y = 0 \text{ m}$) and along the vertical line $x = 0.001, \text{ m}$ for the time step t^2 .



(a) $t^1 = 64.512 \text{ s}$



(b) $t^2 = t^1 + \frac{T}{4} = 64.768 \text{ s}$



(c) $t^3 = t^1 + \frac{T}{2} = 65.024 \text{ s}$

Figure 3.12: Streamlines of a Kármán vortex street in the wake of a cylinder at $Re = 100$ for three consecutive time steps (a)-(c).

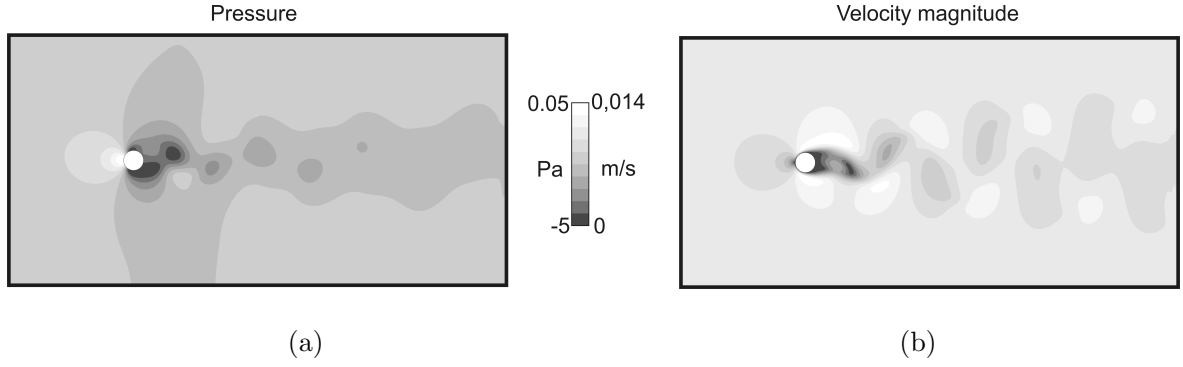


Figure 3.13: Numerical results at t^1 for $Re = 100$ (a) pressure (b) velocity magnitude.

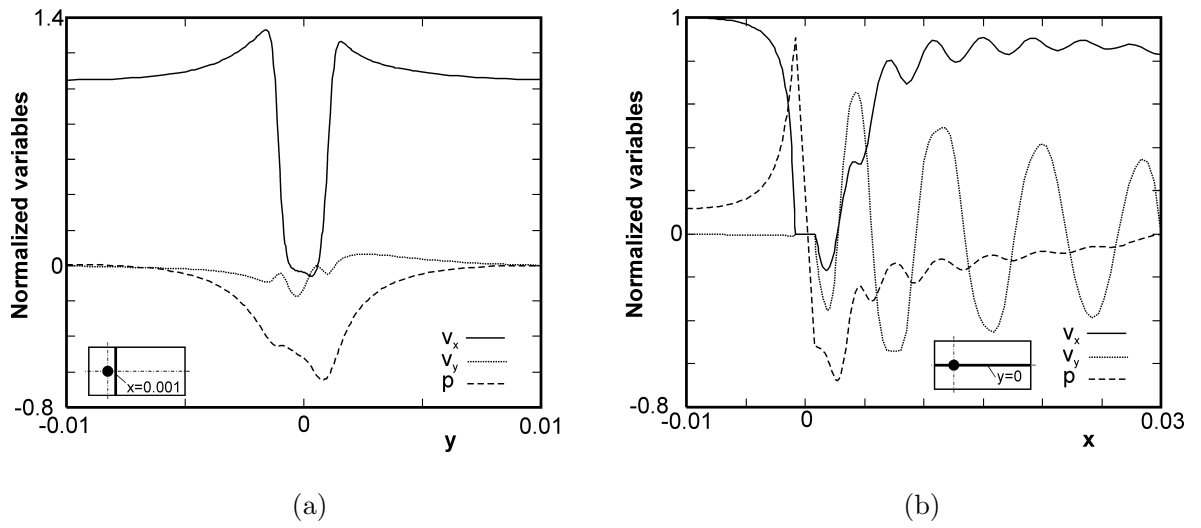


Figure 3.14: Normalized velocity and pressure for $Re = 100$ (a) along the horizontal centerline ($y = 0$ m); (b) along the vertical line ($x = 0.001$ m) for the time step $t^2 = 64.768$ s .